$$R = R_z(\phi) R_y(\theta) R_z(\psi)$$

$$R_z(\phi) = \begin{bmatrix} c\phi & -s\phi & 0 \\ s\phi & c\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} c\theta & 0 & s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix}$$

$$R_z(\psi) = \begin{bmatrix} c\psi & -s\psi & 0 \\ s\psi & c\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\frac{dR}{dt} = \frac{dR_z(\phi)}{dt} R_y(\theta) R_z(\psi) + R_z(\phi) \frac{dR_y(\theta)}{dt} R_z(\psi) + R_z(\phi) R_y(\theta) \frac{dR_z(\psi)}{dt}$$

$$\frac{dR_z(\phi)}{dt} = \dot{\phi}\begin{bmatrix} -s\phi & -c\phi & 0 \\ c\phi & -s\phi & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\frac{dR_y(\theta)}{dt} = \dot{\theta}\begin{bmatrix} -s\theta & 0 & c\theta \\ 0 & 0 & 0 \\ -c\theta & 0 & -s\theta \end{bmatrix}$$

$$\frac{dR_z(\psi)}{dt} = \dot{\psi}\begin{bmatrix} -s\psi & -c\psi & 0 \\ c\psi & -s\psi & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\omega = \dot{\phi}\,\hat{z}_1 + \dot{\theta}\,\hat{y}_2 + \dot{\psi}\,\hat{z}_3$$

$$S(\omega) = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$